

History

When Marie and Parie Curie were engaged in their research on radioactivity, they had very little idea about the dangerous health related effects of radiation

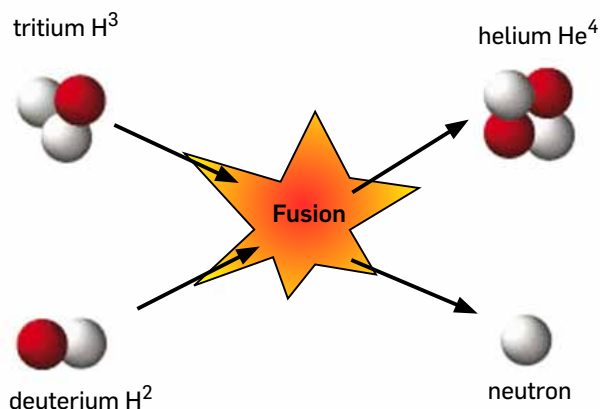
From the labs

nium comes in a mixture of two isotopes - U-235 and U-238. The mixture will contain 99.3 per cent of U-238 and only 0.7 per cent of U-235.

Harnessing uranium

Both U-238 and U-235 are unstable and radioactive, which means that they release nuclear particles in a steady stream till they get turned into lead. However, the half life of U-238 is very large, it stands at around 4.47 billion years. Half life is scientific term for the period of time that it takes for a radioactive material undergoing decay to decrease by half. The half life of U-235 is around 700 million years. That is the prime reason why U-238 occurs in much higher quantities as compared to U-235. Scientists have concluded that around 1.7 billion years ago, the uranium existing on earth must have been naturally enriched, and through the process of nuclear fission, it could have generated lot of heat for around a million years. For use in a nuclear reactor, the uranium needs to be enriched, so that the concentration of U-235 is increased from 0.7 per cent to around 4 to 5 per cent. Even if you have a tonne of U-238, nuclear fission will not take place, and so the process of enrichment is necessary.

During fission an atom of Uranium-235 releases an alpha particle, which consists of two neutrons and two protons bound together. As this alpha particle travels it gets absorbed by the nucleus of another atom, which also becomes unstable and splits into two lighter atoms and two or three new neutrons. The process of capturing of alpha particles and splitting of atoms happens at extremely fast pace and this results in generation of a huge amount of energy. The decay of a single U-235 atom releases approximately 200 MeV (million electron volts). This might not seem like much, but theoretically speaking, a kilogram of Uranium-235 can produce energy, which is equivalent to the energy that can come from 3000 tonnes of coal. However, the waste product of the reactors using U-235 is Plutonium-239, which is itself radioactive and can be used to make a nuclear weapon. The disposal of Plutonium-239 is difficult, as the material can remain radioactive for thousands of years; it has to be buried in deep underground bunkers, which must be leak proof to ensure that there is no seepage into the soil.



It has often been claimed that many governments have focussed on developing Uranium reactors, because they were more interested in the waste by products generated by the Uranium fuel cycle that can be used for making nuclear weapons. In fact there is an

AMKETTE



Share movies, music, photos and internet between two computers.

Connect in a flash!



With Amkette's Flash Link, the high-speed instant sharing device, you can transfer data, share movies, music, photos and even an internet connection between two PCs/laptops. The Flash Link also syncs Outlook mails and lets you access the other computer's CD/ DVD drive. What's more, you can now take back-ups from one computer to another.



Share your friend's CD/DVD Drive



Share one internet connection between the 2 of you



Drag & Drop large files instantly@20mbps



Sync Outlook Mail



Data backup from 1 PC to another

Amkette Flash Link is available at all leading stores across India with 1 year carry in warranty. For further information contact: Amkette House, C-64/4, Okhla Phase-II, New Delhi-110 020 Ph.: 91-11-26389876 / 77, 26388813, E-mail: info@amkette.com or visit: www.amkette.com